



Web Dashboard User Manual

ISP Super Admin & Master Operations

Unified management for Public RVM and Enterprise PecoDrop fleets

INTERNAL ISP USE

ISP Environmental Solutions Pvt. Ltd.

Edition 2026 • Master Portal: isprvm.binishaqsoft.com

1. Manual scope and system boundary

This manual covers the ISP Master Portal used to govern public RVMs, enterprise PecoDrop tenants, users, ESG reporting, rewards, digital signage, fleet configuration and data operations. The portal is a React/Vite interface backed by Node/Express and PostgreSQL.

Public RVM
General-public machines accepting plastic, cans and UBC cartons, with rejects, citizen rewards, mobile-wallet options and public leaderboards.

Enterprise PecoDrop
Corporate dual-display kiosks with QR/keypad authentication and separate plastic, metal/can and weighed-paper paths.

Production authority: PostgreSQL is the sole source of truth. MongoDB is legacy read-only and may only be used to synchronize historical data into PostgreSQL. Never add, edit or delete operational records in MongoDB.

Access roles

Role	Scope	Typical authority
Super Admin	All clients, sites and machines	Tenant approval/deletion, fleet assignment, RBAC, database tools and backups.
ISP operational roles	Modules assigned to the role	Fleet Operator, Analytics Analyst, Support Specialist and field technicians receive limited modules.
Client Admin	One corporate organization	Own fleet, reports, ESG, signage and corporate sub-users.
Corporate Sub-User	Explicit machine subset	Read/operate permitted analytics and machines only.

2. Login, navigation and filters

1. Open the production portal using an approved modern browser.
2. Enter the assigned ISP username and password; never share a Super Admin account.
3. Confirm the identity/role shown in the header before making changes.
4. Use the client and station selectors to set the working scope. Recheck scope before every edit, payout or deletion.
5. Use Logout at the end of the session, especially on shared support computers.

ISP Master Portal — Header: identity • client scope • station filter • theme • logout

Executive Overview

Machine Health & Operations

Digital Signage

Enterprise Accounts

Analytics & Reports

System & Admin

Selected scope: All Sites / Selected Client

Fleet Online
status KPI

Material Intake
volume KPI

Rewards
liability KPI

Charts, alerts and tables inherit the selected tenant/station scope.

Main navigation

Area	Purpose
Executive Overview	Fleet KPIs, material totals, sessions, uptime and operational summary.
Machine Health & Operations	Machine connectivity, telemetry, alerts and site-level status.
Digital Signage	Create, assign, publish and remove kiosk campaigns.
Recycler Community / Rewards	User activity, points, redemptions, rankings and controlled payouts.
Enterprise Accounts	Corporate tenants, departments, employees, branding, admins and fleet bindings.
ESG / Analytics & Reports	Environmental impact, compliance, reconciliation and exportable reporting.
System & Admin	RBAC, machines/configuration tables, database connections and backup/restore.

Theme accessibility: Use the portal’s supported themes. Status and material colors supplement text labels; do not rely on color alone when interpreting alarms.

3. Executive operations and machine health

Daily control-room review

- 1. Select **All Sites** and inspect headline KPIs for offline machines, bin alerts, intake variance and recent activity.
- 2. Open Machine Health & Operations and sort/filter affected machines.
- 3. Confirm machine ID, client, installed location, last heartbeat and reported state.
- 4. Change to a single client or station before investigating detailed events.
- 5. Record the issue and assign it to the correct RVM or PecoDrop technician workflow.

Do not mix machine identities

Machine	Expected material data	Important distinction
RVM	Plastic S/M/L, Can S/M/L, UBC S/M/L and Reject counts	Public sessions, public rewards and citizen identity.
PecoDrop	Plastic units, metal/can units and paper weight	Corporate employee, department and enterprise perks/ESG.

Machine record procedure

- 1. Open the appropriate machine view and search the exact machine ID.
- 2. Verify type, client ownership and location before editing.
- 3. Use configuration tools only for authorized parameters; document the old and new value.
- 4. Confirm the kiosk receives/synchronizes the configuration.
- 5. Validate normal telemetry after the change.

Never “fix” ownership by editing unrelated logs. Machine assignment is managed through tenant/fleet binding. Recycling transactions and physical machine records must remain intact.

Digital signage campaign



- Confirm copyright, brand approval, aspect ratio, schedule and file size before publishing.
- Use destination assignment so RVM and PecoDrop campaigns reach the correct fleet.
- Remove or expire outdated campaigns; do not overwrite evidence needed for an active incident.

4. Enterprise client lifecycle

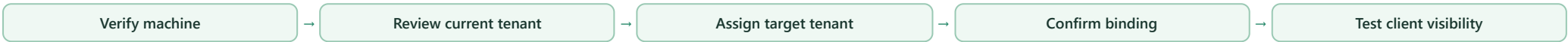
Create and approve a client

1. Open **Enterprise Accounts** and choose Add/Create Organization.
2. Enter official company name, corporate email domain, contact details, monthly targets/budget and approved logo.
3. Review the organization status and approve legitimate onboarding requests.
4. Create the Client Admin account and provide credentials through the approved secure channel.
5. Assign only the intended unassigned/ISP_MASTER machines.
6. Save personalization and verify the client sees only its own organization.

Organization administration

Function	Required checks
Departments	Unique name, responsible manager and correct organization.
Employee roster	Corporate identity, department mapping and privacy authorization; validate bulk-import columns.
Branding	Approved logo, theme color and readable contrast.
Fleet binding	Machine ID, type, installed location and current owner.
Client Admin	Correct org_id, unique username, secure initial credentials and ACTIVE/approved state.

Safe reassignment



Delete a corporate tenant

Super Admin only. Obtain written approval and verify the exact organization. Deletion revokes corporate access and removes organization structures; it must not delete physical machines or recycling history.

1. Export/retain any required business record before deletion.
2. Select the exact tenant; for bulk deletion, recheck every selected card.
3. Confirm bound kiosk IDs and affected administrators.
4. Execute individual or bulk delete and wait for completion.
5. Verify bindings are removed, departments and corporate accounts are purged, and machines return to **ISP_MASTER — ISP Environmental Master (All Sites)**.
6. Confirm recycling logs and physical machine records remain available.

5. Users, RBAC and security

Least-privilege rule

Grant only the modules and machines required for the person’s job. Prefer named accounts over shared credentials. Client Admin and Corporate Sub-User scope must always include the correct organization, with sub-users limited to explicit machine IDs.

Role	Default module pattern
Fleet Operator	Overview and Machines
Analytics Analyst	Overview, Rewards/Analytics, Reports and ESG
Support Specialist	Overview, mobile users, feedback and user support data
PecoDrop/RVM Technician	Overview, Machines and Reports
Client Admin	Own overview/fleet, activity, rewards, ESG, reports, signage and sub-users
Corporate Sub-User	Assigned fleet subset, activity, rewards, ESG and reports; no user/signage administration

Create or change access

1. Confirm requester authority and intended organization/sites.
2. Select the correct role; add only necessary modules and machine IDs.
3. Set a temporary credential through an approved process and require secure change/handling.
4. Test using the affected role, not the Super Admin view.
5. Record approval, effective date and reviewer.

Approval queue

Review pending corporate onboarding requests for corporate email/domain, organization, contact and business authorization. Approve legitimate requests or reject with an auditable reason.

Immediate response: Suspend/revoke access for leavers, compromised accounts or unauthorized scope. Do not delete audit evidence while investigating.

6. Rewards, users and reconciliation

Recycler investigation

1. Open Recycler Community and search by the permitted user identifier.
2. Compare recycling sessions, earned points, redeemed points and current balance.
3. Review voucher/redemption code, status, category and timestamp.
4. Correct source data only through an approved PostgreSQL-backed application workflow.

Issue payout or voucher

1. Select the correct recycler and verify destination mobile/wallet details.
2. Choose EasyPaisa, JazzCash, airtime, merchant voucher or Raast as applicable.
3. Confirm PKR amount and required points; the standard displayed rule is 5 points = PKR 1.
4. Do not enable overdraft without explicit financial authorization.
5. Submit once, retain the returned reference and verify the redemption record.

Vouch365 status: The current codebase contains a Vouch365-labelled WebView/link and locally generated voucher references, but no authenticated external Vouch365 issuance API. Do not represent a locally generated code as externally funded or delivered until a provider response is implemented and verified.

Reporting

- Set client/site/date filters before generating a report.
- Reconcile material totals, sessions, rewards, voucher liability and ESG values.
- Export only the minimum required personal data and store it according to ISP policy.
- State the report scope, timezone and generation date when sharing externally.

7. Database connections, synchronization and backups

Database Connections

- Use PostgreSQL as production authority for all additions, updates and deletions.
- Use MongoDB only as a legacy read source for controlled historical synchronization into PostgreSQL.
- Review health/test results before switching or synchronizing.
- Never sync PostgreSQL changes back into MongoDB.

Backup procedure

1. Confirm active production database and sufficient protected storage.
2. Create a backup and wait for successful completion.
3. Record filename, timestamp, size and operator.
4. Download/store according to retention policy and restrict access.
5. Periodically test restoration in a non-production environment.

Restore procedure

High-impact operation: Obtain change approval, define recovery point and maintenance window, notify users, take a fresh pre-restore backup, then restore and validate.

1. Select the verified backup by timestamp and integrity.
2. Execute restore once; do not interrupt the server.
3. Validate login, tenant scope, machine inventory, recent sessions, rewards and health endpoints.
4. Record outcome and rollback decision.

Incident checklist

☐	Capture time, user, selected client/site and affected module.
☐	Record request/error message without exposing tokens or passwords.
☐	Check portal health, PostgreSQL and machine connectivity.
☐	Preserve logs and audit records.
☐	Escalate financial, privacy or cross-tenant exposure immediately.

8. ISP operational checklists

Start of shift

☐	Portal health and PostgreSQL connection normal.
☐	Offline machines/bin alerts reviewed.
☐	Pending tenant/user approvals reviewed.
☐	Failed signage/config pushes reviewed.
☐	Reward/redemption exceptions reviewed.

End of shift

☐	Critical incidents handed over with machine/client IDs.
☐	High-impact changes and deletions documented.
☐	Backup status checked according to schedule.
☐	Exports containing personal data secured or removed.
☐	Portal session logged out.

Change record

Date/time		Operator	
Client/site		Machine(s)	
Change			
Validation			
Approval/reference			

Visual Guide 1 — Secure login

Actual portal capture • 2026


ISP Environmental Solutions ✨
SMART RECYCLING PORTAL

🔑 **USERNAME OR EMAIL**
Enter your username or email

🔑 **PASSWORD**
Enter your password

☒ Remember me [Forgot password?](#)

SIGN IN

© Authorized Administrative Personnel Only

Use an individual account. Never include passwords in screenshots or support tickets.

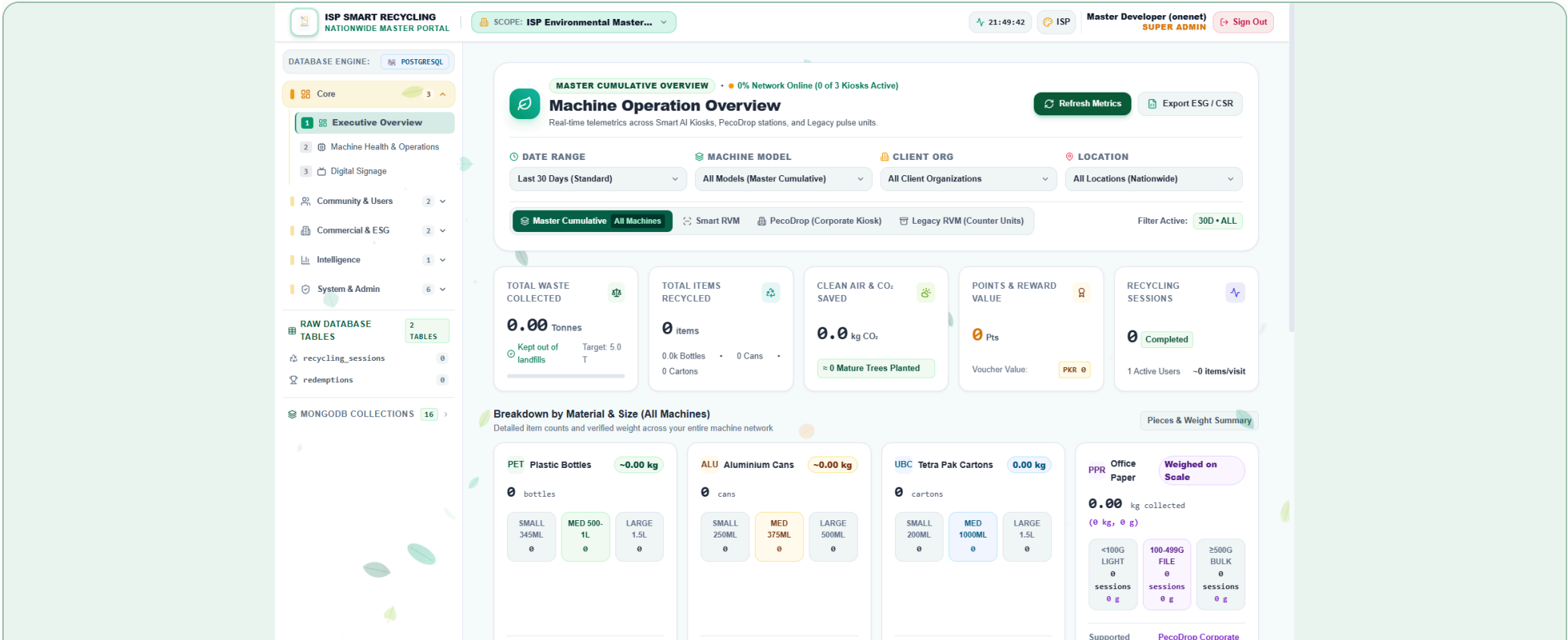
Identity
Approved ISP username.

Password
Protect shared workstations.

Session
Remember Me only on private devices.

Visual Guide 2 — Executive overview

Master cumulative fleet view



Confirm scope, date range, model, client and location before interpreting KPIs.

Scope
Sets tenant context.

KPIs
Waste, impact, rewards and sessions.

Materials
PET, cans, UBC and paper.

Visual Guide 3 — Machine health

ISP SMART RECYCLING
NATIONWIDE MASTER PORTAL

SCOPE: ISP Environmental Master...

21:49:45 ISP Master Developer (onenet)
SUPER ADMIN Sign Out

DATABASE ENGINE: POSTGRES

Core 3

1 Executive Overview

2 Machine Health & Operations

3 Digital Signage

Community & Users 2

Commercial & ESG 2

Intelligence 1

System & Admin 6

RAW DATABASE TABLES 2

recycling_sessions 0

redemptions 0

MONGODB COLLECTIONS 16

HARDWARE MONITORING | Machine Network Telemetry

Refresh Heartbeat Dispatch Field Tech Fleet GPS Map ON

Machine Health & Operations

CONNECTED MACHINES 3 deployed units

0 Smart RVM - 2 PecoDrop - 1 Legacy

ONLINE & ACTIVE 0 Synced

Awaiting network heartbeat

AWAITING HEARTBEAT 3 units

Field standby or local network offline

MAINTENANCE FLAGS 3 Needs Attention

Hopper bins >85% full or sensor alerts

All Machines (3) Smart RVM (0) PecoDrop Scale Units (2) Legacy Units (1) Action Needed (3)

Sort by: Recent Heartbeat Ping

PECO-01 PecoDrop Offline

PecoDrop Station (Indoor H...
ISP Metro Stree...

Client: Metro Mall RWP

3-Bin Capacity Load Scale: 99.8%

Plastic 58% Metal 42% Paper 11.4 kg

Recycling Sessions: 0 completed

Bottles Processed: 0 bottles

Cans Processed: 0 cans

Paper Processed: 14.2 kg

Points Issued: 0 pts

PUBLIC IP 127.0.0.1 LOCAL IP 127.0.0.1

Legacy Counter Offline

RVM Old Legacy K...
ISP Metro Stree...

Client: Metro Cash & Carry

Relay Counter Bridge Queue Synced

1420 Hardware Pulses recorded - 0 backlog

Plastic Bin Overfilling Dispatch Tech

Chamber reached 85% physical capacity

Recycling Sessions: 0 completed

Bottles Processed: 0 bottles

Cans Processed: 0 cans

PECO-02 PecoDrop Offline

PecoDrop Station (Indoor H...
Metro Mall RWP - Food ...

Client: Metro Mall RWP

3-Bin Capacity Load Scale: 99.8%

Plastic 65% Metal 50% Paper 14.8 kg

Recycling Sessions: 0 completed

Bottles Processed: 0 bottles

Cans Processed: 0 cans

Paper Processed: 14.2 kg

Points Issued: 0 pts

PUBLIC IP 127.0.0.1 LOCAL IP 127.0.0.1

Search the exact machine ID and preserve selected scope while investigating.

Filter first

Narrow the fleet.

Identity

Verify type, owner and heartbeat.

Escalate

Record evidence.

Visual Guide 4 — Digital signage

ISP SMART RECYCLING
NATIONWIDE MASTER PORTAL

SCOPE: ISP Environmental Master...

21:49:48 ISP Master Developer (onenet)
SUPER ADMIN Sign Out

DATABASE ENGINE: POSTGRESQL

Core 3

Executive Overview

Machine Health & Operations

Digital Signage

Community & Users 2

Commercial & ESG 2

Intelligence 1

System & Admin 6

RAW DATABASE TABLES 2

recycling_sessions 0

redemptions 0

MONGODB COLLECTIONS 16

DIGITAL SIGNAGE HUB Machine Displays Live Sync

Screen Ad & Video Manager

Sync All Displays Upload New Video Ad AD

ACTIVE MEDIA LIBRARY 3 video assets
2 in rotation • 1 targeted single unit

ACTIVE SCREENS 8 100% Synced
5 RVM Headers • 3 PecoDrop Screens

SIGNAGE STORAGE 48.2 MB Used
Optimized MP4 video format (H.264)

PLAYBACK MODE Seamless Loop
Autoplays during standby

Target Display Scope & Machine Filter
Choose which location or individual machine display you want to manage

View All Video Campaigns

Target Network Scope Select Facility / Location Hardware Screen Profile

Global Fleet (All 8 Machines) All Locations (Nationwide) All Display Orientations

Configured Video Ad Campaigns
Manage video assignments, sequencing, and live machine routing

Showing 3 Campaigns

16:9 Landscape PecoDrop Exclusive
PECO Corporate Green Journey Active Loop
0:30

16:9 Header Display Public RVM
Pepsi Recycle & Earn PKR 200 Active Loop
0:15

Single Machine Unit Campus Specific
University Plastic Bottle Drive Single Spot
0:45

Create approved media, assign destinations and verify kiosk synchronization.

Library
Review dates and status.

Destinations
Separate RVM/PecoDrop.

Verify
Confirm physical playback.

Visual Guide 5 — Enterprise accounts

ISP SMART RECYCLING
NATIONWIDE MASTER PORTAL

SCOPE: ISP Environmental Master...

21:50:00 ISP Master Developer (onenet)
SUPER ADMIN Sign Out

DATABASE ENGINE: POSTGRESQL

Core 3

Community & Users 2

Commercial & ESG 2

Enterprise Accounts 6

ESG & Carbon Impact 7

Intelligence 1

System & Admin 6

RAW DATABASE TABLES 2 TABLES

recycling_sessions 0

redemptions 0

MONGODB COLLECTIONS 16

MULTI-TENANT B2B ENTERPRISE ENGINE PET Bottles Cans Paper

Enterprise Corporate Clients & ESG

Manage corporate client tenants, automated domain routing (@company.com), department leaderboards, touchless staff claims, and multi-material recycling audit compliance.

User Guide (PDF / ریدو) Add Enterprise Client

CORPORATE CLIE... 4 Active

ENROLLED STAFF 0

PAPER RECYCLED 0 kg

BOTTLES & CANS 0 0 PET / 0 Cans

TREES SAVED (ESG) 0

Search enterprise clients by company name or domain (@bankalfalah.com)...

Select All Showing 4 of 4 Clients

Bank Alfalah Limited @bankalfalah.com

Bottles 0 Cans 0 Paper 0 kg

Monthly ESG Target 0 / 2,500 kg

Staff 0 Depts 3 ESG Points 0

Bound Fleet: 0 Bound Kiosks

Client Admin: Not Provisioned

Personalize Fleet Depts & Roster

Engro Corporation @engro.com

Bottles 0 Cans 0 Paper 0 kg

Monthly ESG Target 0 / 2,000 kg

Staff 0 Depts 2 ESG Points 0

Bound Fleet: 0 Bound Kiosks

Client Admin: @engro_admin

Personalize Fleet Depts & Roster

Metro Cash & Carry @metro.pk

Bottles 0 Cans 0 Paper 0 kg

Monthly ESG Target 0 / 1,800 kg

Staff 0 Depts 0 ESG Points 0

Bound Fleet: 0 Bound Kiosks

Client Admin: Not Provisioned

Personalize Fleet Depts & Roster

Review status, ESG targets, departments, fleet and Client Admin provisioning.

Add client
Validate domain and logo.

Fleet
Assign verified machines.

Deletion
Return fleet; retain logs.

Visual Guide 6 — Rewards

ISP SMART RECYCLING
NATIONWIDE MASTER PORTAL

SCOPE: ISP Environmental Master...

21:49:56 ISP Master Developer (onenet)
SUPER ADMIN Sign Out

DATABASE ENGINE: POSTGRESQL

Core 3

Community & Users 2

Recycler Community 4

Rewards & Leaderboards 5

Commercial & ESG 2

Intelligence 1

System & Admin 6

RAW DATABASE TABLES 2

recycling_sessions 0

redemptions 0

MONGODB COLLECTIONS 16

COMMUNITY GAMIFICATION | Top Recyclers Live Ranks

Green Champions & Recycling Leaderboard

Refresh Rankings Issue Payout / Voucher ON

Search by registered citizen name, contact number, or corporate client (e.g. Tariq, Engro, A) All Clients & Citizens All Time This Month Corporate Units

SILVER RECYCLER 2 No recycler data
No matching activity for this filter

ALL-TIME CHAMPION 1 Gold Tier No recycler data
No matching activity for this filter

BRONZE RECYCLER 3 No recycler data
No matching activity for this filter

Recycled Volume: 0 Items (PET & Cans)
Earned Points: 0 Pts
Reward Voucher Value: PKR 0

Recycled Volume: 0 Bottles Diverted
Cumulative Balance: 0 Pts
Equivalent Voucher Value: PKR 0

Recycled Volume: 0 Bottles & UBC Cartons
Earned Points: 0 Pts
Reward Voucher Value: PKR 0

Top 8 Recyclers Point Intake Distribution
Audited loyalty points generated per verified active recycler

Metric Scale: Points in Thousands (k)

Engagement Motivation Active

Campaign milestones to retain active recyclers

1,000 Points = PKR 200
Users reach minimum cash-out threshold at 1,000 points. Keep public kiosk push notifications active!

PecoDrop Team Competitions
Corporate campus employees earn bonus reward badges for confidential document drops over 1 kg.

Top Recycler of the Month
The #1 monthly contributor receives an additional PKR 2,500 shopping voucher sponsored by FMCG partners

Validate user, points, destination, method and approval before payout.

Balance

Earned minus redeemed.

Provider

Local code is not fulfilment proof.

Audit

Retain reference.

Visual Guide 7 — Access and security

ISP SMART RECYCLING
NATIONWIDE MASTER PORTAL

SCOPE: ISP Environmental Master...

21:50:12 ISP Master Developer (onenet)
SUPER ADMIN Sign Out

DATABASE ENGINE: POSTGRES

Core 3

Community & Users 2

Commercial & ESG 2

Intelligence 1

System & Admin 6

Access & Security (RBAC)

Smart Recycling Machines 10

Smart Recycling Configurations 11

Users 12

Database Connections 13

Backups & Restore 14

RAW DATABASE TABLES 2

recycling_sessions 0

redemptions 0

MONGODB COLLECTIONS 16

ENTERPRISE SECURITY & RBAC

User Accounts & Role Permissions

Manage user info, passwords, role assignments, and customizable Module Permissions Matrix

User Accounts Role Access Matrix Create New User

Active System Accounts Directory (5)

USER ACCOUNT	ASSIGNED ROLE	ASSIGNED SMART RECYCLING MACHINES SCOPE	ACCESSIBLE MODULES	STATUS	ACTIONS
testingrvm testingrvm@gmail.com	RVM Fleet Operator	All Recycling Fleet (*)	MACHINE HEALTH & OPERATIONS EXECUTIVE OVERVIEW	ACTIVE	Edit Info
Tariq Mehmood - Head of CSR csr@engro.com	Client Admin (Site Tenant)	PECO-LHR-01 RVM-LHR-01	EXECUTIVE OVERVIEW REWARDS & LEADERBOARDS ESG & CARBON IMPACT ANALYTICS & REPORTS DIGITAL SIGNAGE MACHINE HEALTH & OPERATIONS	ACTIVE	Edit Info
Ahmed Khan - Lahore Site Operator ahmed.lahore@engro.com	corporate_sub_user	PECO-LHR-01	No menus assigned	ACTIVE	Edit Info
Illegal User illegal_user@rvm-dash.io	corporate_sub_user	PECO-01	No menus assigned	ACTIVE	Edit Info
Bilal Aqeel bilalaaqeel@gmail.com	Super Admin / Master Dev	All Recycling Fleet (*)	EXECUTIVE OVERVIEW REWARDS & LEADERBOARDS MACHINE HEALTH & OPERATIONS	ACTIVE	Edit Info

Assign the smallest role, module set and machine scope required.

Accounts
Use named identities.

Modules
Limit by job.

Lifecycle
Suspend leavers.

Visual Guide 8 — Backup and restore

High-impact operation

ISP SMART RECYCLING
NATIONWIDE MASTER PORTAL

SCOPE: ISP Environmental Master...

21:58:15 ISP Master Developer (onenet)
SUPER ADMIN Sign Out

DATABASE ENGINE: PostgreSQL

Core 3
Community & Users 2
Commercial & ESG 2
Intelligence 1
System & Admin 6
Access & Security (RBAC) 9
Smart Recycling Machines 10
Smart Recycling Configurations 11
Users 12
Database Connections 13
Backups & Restore 14

RAW DATABASE TABLES 2
recycling_sessions 0
redemptions 0
MONGODB COLLECTIONS 16

ISP Smart Recycling Hub
PG Host: 127.0.0.1:5432

DATABASE MANAGEMENT
Database Backup & Restore Utility (PostgreSQL)
Connected Engine: PostgreSQL • Target DB: rvmpg • Host: 127.0.0.1:5432
Create & Download Backup Direct JSON Export

Restore JSON Backup into Connected Database (rvmpg)
Select or drop a 'json' backup file to restore into database rvmpg
Will populate collections: recycling_sessions, userprofile, feedbacks, binfullnotifications, etc.
Browse JSON Backup File

Stored Backup Snapshots (0)
Backup File Name Created Date File Size Actions
No backup snapshots stored yet. Click "Create & Download Backup" above.

Restore only with approval, recovery point, window and validation plan.

Before
Fresh backup.

During
Do not interrupt.

After
Validate all services.